

Behavioral Economics

FALL 2018

Instructor: Professor Alex Imas

Time and Location: POS 146, Tuesdays and Thursdays, 10:30-11:50 a.m.

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Outline and Objectives

OVERVIEW AND GOALS

This course will be taught seminar-style, with lectures on the behavioral economics literature studying decision-making and strategic interactions and interactive presentations of recent and original research. One of the main goals of the course is to provide students with hands-on experience on how to design and carry out their own experiments. To this end, active experimentation using both traditional and cutting-edge research methods will constitute a significant portion of the class.

The lectures will explore how psychology and economics have been integrated to form behavioral economics, giving researchers a unique set of tools to study how people respond to risk, make choices over time, and interact with others. We will then do an overview of how behavioral economics has been used to test game theory of strategic interactions and how people learn about their environment and the actions of others. The last sections of the class will briefly review work on behavioral finance and emotions.

Students will have an opportunity to critically discuss existing behavioral economics research as well as to present their own research ideas. Everyone will have an opportunity to both participate and run their own experiments.

PREREQUISITES

Students should be comfortable with basic economic principles such as supply and demand, utility functions, and choice under risk.

We will not be using that much math in the class, but familiarity with certain theoretical constructs is required in order to understand why experiments are used to study them.

FORMAT

The course will be taught in a seminar format. Students are expected to interact, comment and challenge the presenter. The class is organized into modules, with each module composed of two to four classes focused on one of the topics listed below. The first classes in a module will introduce the seminal papers in the literature, one class will be devoted to recent developments on the topic, and the last class of each module will feature student presentations on the topic. Students will both participate in experiments and conduct their own in line with the theme of each module.

REQUIREMENTS AND GRADING

Since a major goal of the class is the development of research skills, there will be no in-class midterms or final. Presentations, participation and the take-home assignments will be used to determine grading. Students will have an opportunity to present their own ideas and existing working papers of others throughout the semester. In addition, a final paper will be due at the end of the semester.

Participation

Given the seminar format of this course, students are expected to play an active role in class discussions. Students should plan on preparing 2-3 questions on each of the required readings to be discussed. The last 10-20 minutes of each class will be devoted to active discussions of the papers, so bring your questions to class!

The beginning of each lecture class will start with a short 2-3 question quiz on the readings. Scores on these quizzes will count towards your participation grade in the class.

In addition, students will be assigned to present a working or recently published paper from the reading list on each topic. The presentations should be around 30 minutes, and designed to facilitate discussion. The presenters are encouraged to incorporate their own thoughts and opinions about the paper.

Idea Briefs

For each topic covered, students will write a short idea brief (1-2 pages) focusing on a research idea related to the topic. The idea briefs should focus on the big picture questions that the research aims to answer, with a focus on the motivation and the experimental design used to explore it. For each module, ten students will be chosen to give short (5-10 minute) presentations of their idea brief on the day that they are due.

Original Experiments

Throughout the class we will be using the Mechanical Turk crowdsourcing platform to conduct experiments in real-time. Students will be expected to submit one original experiment for each module that will be voted on (double-blind) by the class. The winning experiment will be carried out and the results discussed. The winner will receive extra credit, equivalent to one idea brief.

Final Project

The final project is a paper (10-20 pages) on an original research topic of the students choice. The paper should be structured as a standard scientific article, with an introduction, literature review, procedures, results, discussion and conclusion. Actual data is not required. If no data was collected, the results section should contain a detailed plan on how such data would be analyzed. Students can work together. Each student or group will make a brief presentation (20 minutes) of their final project.

Grading

The grade will be based 40% on participation, 20% on idea briefs, 10% on original experiments and 30% on the final project.

Depending on the distribution on points the class can be curved up (never down). Hence, if you have over 90% of the possible points you are going to have an A regardless of how everybody else does. Letter grades are in principle assigned according to the percentage of total points according to the following scale:

100-90	A
89- 80	B
79-70	C
69-60	D
< 60	R

TEXT

Unfortunately, there is no standard behavioral economics textbook. So, most of the material will come from the lecture. I will post a version of the lecture slides on blackboard the night before each class. The slides will provide a detailed outline of lecture, however they should not be considered exhaustive.

Required and optional reading will be posted to blackboard and the course website. The following text is encouraged:

Camerer, C. F., Loewenstein, G., and Rabin, M. *Advances in Behavioral Economics*, New York, Princeton University Press, 2004.

Administrative Issues and Policies

EMAIL

The best way to reach me is via email. I will do my best to answer emails as soon as I receive them, but it is not always possible. Include the TA on the email to speed up the process. In addition, I will send out class information to the email you have on file with the university (through Canvas), so make sure that email is correct and course emails do not go to the spam folder.

ASSIGNMENT POLICY

You must turn in all the assignments by the due date *during the first 10 minutes of class* to receive full credit. After that, you may turn an assignment until the end of the day the assignment is due, with a penalty of 25% of the total possible points. Under no circumstances you may turn any assignment later than the day it is due.

Classroom Etiquette

ARRIVE ON TIME

Please arrive on time at the start of class. We will often be running experiments in the class, with you as the participants. If you arrive late, you'll miss the instructions and will not be able to participate.

NO SCREENS POLICY

Active attention and participation are really important for an interactive course such as behavioral economics. For that reason, during class do not use laptops, tablets, cell phones, or whatever new screen gets invented to distract you from the people around you. If you require any accommodations for this course, please see me. I am committed to ensuring the full participation of all students.

ATTENDANCE

You are expected to attend all of the classes. In order to run classroom experiments, we need to know the number of students who are going to be in the class. If you have a legitimate conflict, please let me or the TA know ahead of time.

Academic Integrity

Academic integrity is a serious issue. All things considered, a marginally better grade is not every worth engaging in academic dishonesty (copying, cheating, plagiarizing). Many of the assignments are collaborative by nature, and I encourage you to work with your classmates on them. However, each student should turn in their own assignment and clearly indicate who they had worked with. Copying, cheating, or plagiarizing are going to be penalized.

Take Care of Yourself

Do your best to maintain a healthy lifestyle this semester by eating well, exercising, avoiding drugs and alcohol, getting enough sleep and taking some time to relax. This will help you achieve your goals and cope with stress.

All of us benefit from support during times of struggle. You are not alone. There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. Asking for support sooner rather than later is often helpful.

If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support. Counseling and Psychological Services (CaPS) is here to help: call 412-268-2922 and visit their website at <http://www.cmu.edu/counseling/>. Consider reaching out to a friend, faculty or family member you trust for help getting connected to the support that can help.

If you have questions about this or your coursework, please let me know.

Approximate Course Outline

This is an ambitious list of topics for the semester. We might not get through all of it if some topics take more time than expected, and that is OK.

Required readings are in **bold**. Readings not in bold are meant to supplement the lectures.

August 28 - Introduction

Assignments Due: None

Behavioral and Experimental Economics

- Loewenstein, G. (1999). Experimental Economics from the Vantage-Point of Behavioral Economics. *Economic Journal*, 109, 25-34.

Methods of Behavioral Economics

- Hertwig, R. and Ortmann, A. (2001). Experimental Practices in Economics: A Methodological Challenge for Psychologists? *The Behavioral and Brain Sciences*, 24, 383-403.

August 30 - Risk Preferences and Mental Accounting: Foundations 1

Assignments Due: Required Readings

Expected Utility and Prospect Theory

- **Kahneman, D. and Tversky, A. (1979). Prospect Theory: An Analysis of Decision Under Risk. *Econometrica*, 47, 263-292.**

September 4 - Risk Preferences and Mental Accounting: Foundations 2

Assignments Due: Required Readings

Endowment Effect

- **Kahneman, D., Knetsch, J. and Thaler, R. (1991). Anomalies: The Endowment Effect, Loss Aversion, and Status Quo Bias. *Journal of Economic Perspectives*, 5, 193-2006.**

Mental Accounting

- Thaler, R. (1999). Mental Accounting Matters. *Journal of Behavioral Decision Making*, 12, 183-206.

History-Dependent Risk Preferences

- Thaler, R. and Johnson, E. (1990). Gambling with the House Money and Trying to Break Even: The Effects of Prior Outcomes on Risky Choice. *Management Science*, 36, 643-660.

Ambiguity Aversion

- Ellsberg, D. (1961). Risk, Ambiguity, and the Savage Axioms. *Quarterly Journal of Economics*

September 6 - Risk Preferences and Mental Accounting: Advances

Assignments Due: Required Readings, Experimental Idea

- Barberis, N. (2013). Thirty Years of Prospect Theory: A Review and Assessment. *Journal of Economic Perspectives*, 27, 173-196.

Presentation

- Abeler, A. and Marklein, F. (2016) Fungibility, Labels and Consumption. *Journal of the European Economic Association*, 15, 99-127.

Presentation: Abby Kuznick, Madeline Moon, Taylor Dirks, Colin McNeil, Scott Steinberg, Benjamin Stone, Julia Benson

September 11 - Risk Preferences and Mental Accounting: Advances

Assignments Due: Idea briefs

September 13 - Time Preferences: Foundations 1

Assignments Due: Required Readings

- Loewenstein, G. and Thaler, R. (1989). Anomalies: Intertemporal Choice. *Journal of Economic Perspectives*, 3, 181-193.

September 18 - Time Preferences: Foundations 2

Assignments Due: Required Readings

- Ariely, D. and Wertenbroch, K. (2002) Procrastination, Deadlines, and Performance: Self-Control by Precommitment, *Psychological Science*, Vol. 13, 219-224.

September 20 - Time Preferences: Advances

Assignments Due: Required Readings, Experimental Idea

- Read, D., Frederick, S., Orsel, B. and Rahman, J. (2005). Four Score and Seven Years from Now: The Date/Delay Effect in Temporal Discounting. *Management Science*, 51, 1326-1335.

Presentation

- Imas, A., Kuhn, M. and Mironova, V. Waiting to Choose, mimeo.

Presentation: Alex Miller, Alyssa Kloehn, Maury Jackson, Benjamin Huber, Joshua Cagan, Qiutong Dong, Emily Bippus

September 25 - Time Preferences: Advances

Assignments Due: Idea briefs

September 27 - Social Preferences: Foundations

Assignments Due: Required Readings

Social Preferences (Global)

- **Loewenstein, G., Thompson, L. and Bazerman, M. (1989). Social Utility and Decision Making in Interpersonal Contexts. Journal of Personality and Social Psychology, 57, 426-441.**

Reciprocity and Fairness

- **Camerer, C. and Thaler, R. (1995). Anomalies: Ultimatums, Dictators and Manners. Journal of Economic Perspectives, 9, 209-219.**
- Dawes, R. M. and Thaler, R. (1988). Anomalies: Cooperation. Journal of Economic Perspectives, 2, 187-197.

October 2 - Social Preferences: Advances

Assignments Due: Required Readings, Experimental Ideas

- Dana, J., Weber, R. A. and Kuang, J. X. (2007). Exploiting Moral Wiggle Room: Experiments Demonstrating an Illusory Preference for Fairness. Economic Theory, 33, 67-80.
- Dana, J., Cain, D. and Dawes, R. (2006). What You Dont Know Wont Hurt Me: Costly (but Quiet) Exit in Dictator Games. Organizational Behavior and Human Decision Processes, 100, 193-201.

Presentation

- **Exley, C. (2015). Excusing Selfishness in Charitable Giving: The Role of Risk. Review of Economic Studies, 83, 587-628.**

Presentation: Madeleine Bomberg, Rui Hu, Aden Halpern, Timothy Kusuma, Sophia Angel, Jung Inn Yang

October 4 - Social Preferences: Advances

Assignments Due: Idea briefs

October 9 - Strategic Interactions and Markets: Foundations 1

Assignments Due: Required Readings

Markets

- **Smith, V. L. (1962). An Experimental Study of Competitive Market Behavior. *Journal of Political Economy*, 70, 111-137.**
- Kahneman, D., Knetsch, J. and Thaler, R. (1986). Fairness as a Constraint on Profit Seeking: Entitlements in the Market. *American Economic Review*, 76, 728-741.

October 11 - Strategic Interactions and Markets: Foundations 2

Assignments Due: Required Readings

Game Theory

- **McKelvey, R. and Palfrey, T. (1992). An Experimental Study of the Centipede Game. *Econometrica*, 60, 803-836.**

Behavioral Game Theory

- **Nagel, R. (1995). Unraveling in Guessing Games: An Experimental Study. *American Economic Review*, 85, 1313-1326.**

October 16 - Strategic Interactions and Markets: Advances

Assignments Due: Required Readings, Experimental Ideas

- Dufwenberg, M., Lindqvist, T. and Moore, E. (2005). Bubbles and Experience: An Experiment. *American Economic Review*, 95, 1731-1737.

Presentation

- **Falk, A. and Szech, N. (2013). Morals and Markets. *Science*, 340, 707-711.**

Presentation: Bridget Mensah, Elizabeth Finneran, Jennifer Mayberger, Gregory Volynsky, Bolade Fatade, Kevin Kelly, Lauren Song

October 18 - Strategic Interactions and Markets: Advances

Assignments Due: Idea briefs

October 23 - Learning and Persuasion: Foundations

Assignments Due: Required Readings

Communication

- **Schelling, T. (1957). Bargaining, Communication, and Limited War. Conflict Resolution, 1, 19-36.**

Herding and Information Cascades

- Anderson, L. and Holt, C. (1997). Information Cascades in the Laboratory. American Economic Review, 87, 847-862.

October 25 - No Class

October 30 - Putting Behavioral Economics to Work 1

Assignments Due: None

November 1 - Learning and Persuasion: Advances

Assignments Due: Required Readings, Experimental Ideas

- Gneezy, U. (2005). Deception: The Role of Consequences. American Economic Review, 95, 384-394.

Presentation

- **Enke, B. What You See Is All There Is. Mimeo.**

Presentation: Heitho Shipp, Darian Cohen, Weijia Mi, Gregory Bellwoar, Ryan Casey, Amer Ahmad

November 6 - Learning and Persuasion: Advances

Assignments Due: Idea briefs

November 8 - Incentives and Labor Markets: Foundations

Assignments Due: Required Readings

Crowding out and psyching out

- **Gneezy, U., Meier, S. and Rey-Biel, P. (2011). When and Why Incentives (Dont) Work to Modify Behavior. Journal of Economic Perspectives, 25, 191-210.**
- Ariely, D., Gneezy, U., Loewenstein, G. and Mazar, N. (2008). Large Stakes and Big Mistakes. Review of Economic Studies, 76, 451-469.

Labor Markets

- Fehr, E., Kirchler, A., Weichbold, A. and Gächter, S. (1998) When Social Norms Overpower Competition: Gift Exchange in Experimental Labor Markets. Journal of Labor Economics, 16, 324-351.

November 13 - Incentives and Labor Markets: Advances

Assignments Due: Required Readings, Experimental Ideas

- Gneezy, A., Gneezy, U., Reinder, G. and Nelson, L. (2012). Pay-What-You-Want, Identity, and Self-Signaling in Markets. *PNAS*, 109, 7236-7240.
- Imas, A. (2013). Working for the Warm Glow: On the Benefits and Limits of Prosocial Incentives. *Journal of Public Economics*, forthcoming.

Presentation

- **Gneezy, A., Gneezy, U., Nelson, L. and Brown, A. (2010). Shared Social Responsibility: A Field Experiment in Pay-What-You-Want Pricing and Charitable Giving. *Science*, 329, 325-327.**

Presentation: Adrian Nakic, William Wen-Yan Wang, Jennifer Flores, Megan Jaeb, Zachary Goldstein, Jin Fang

November 15 - Incentives and Labor Markets: Advances

Assignments Due: Idea briefs

November 20 and 22, Thanksgiving Holiday

November 27 - Behavioral Finance and Emotions

Assignments Due: Required Readings

Myopic Loss Aversion

- **Gneezy, U. and Potters, J. (1997). An Experiment on Risk Taking and Evaluation Periods. *Quarterly Journal of Economics*, 112, 631-645.**

Disposition Effect

- Shefrin, H. and Statman, M. (1985). The Disposition to Sell Winners Too Early and Ride Losers Too Long: Theory and Evidence. *Journal of Finance*, 40, 777-790.

Risk and Affect

- **Shiv, B., Loewenstein, G., Bechara, A., Demasio, H. and Demasio, A. (2005). Investment Behavior and the Negative Side of Emotion. *Psychological Science*, 16, 435-439.**

Emotions and Empathy Gaps

- Loewenstein, G. (1996). Out of Control: Visceral Influences on Behavior. *Organizational Behavior and Human Decision Processes*, 65, 272-292.

November 29 - Putting Behavioral Economics to Work 2

Assignments Due: None

December 4 and 6

Final Presentations